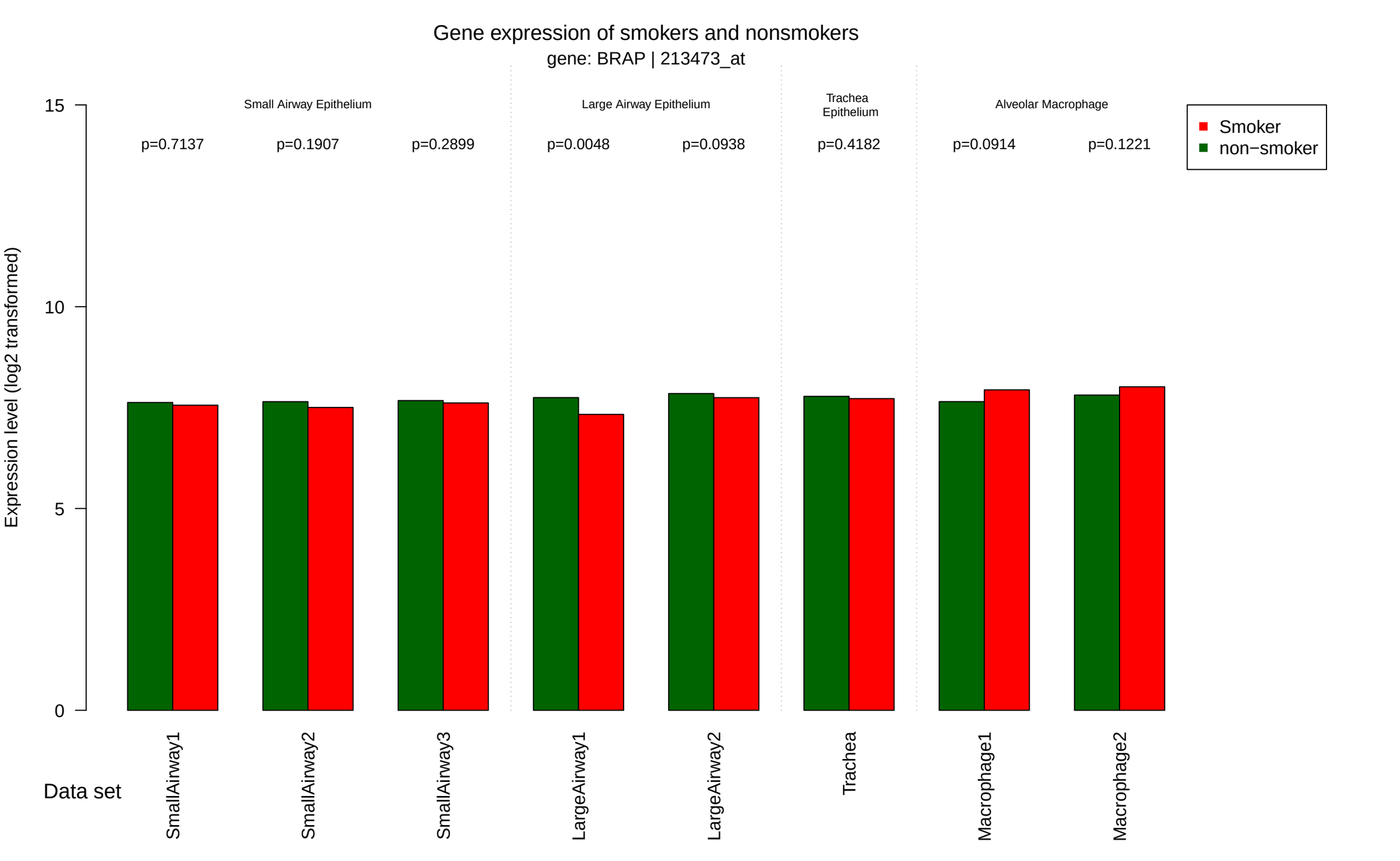
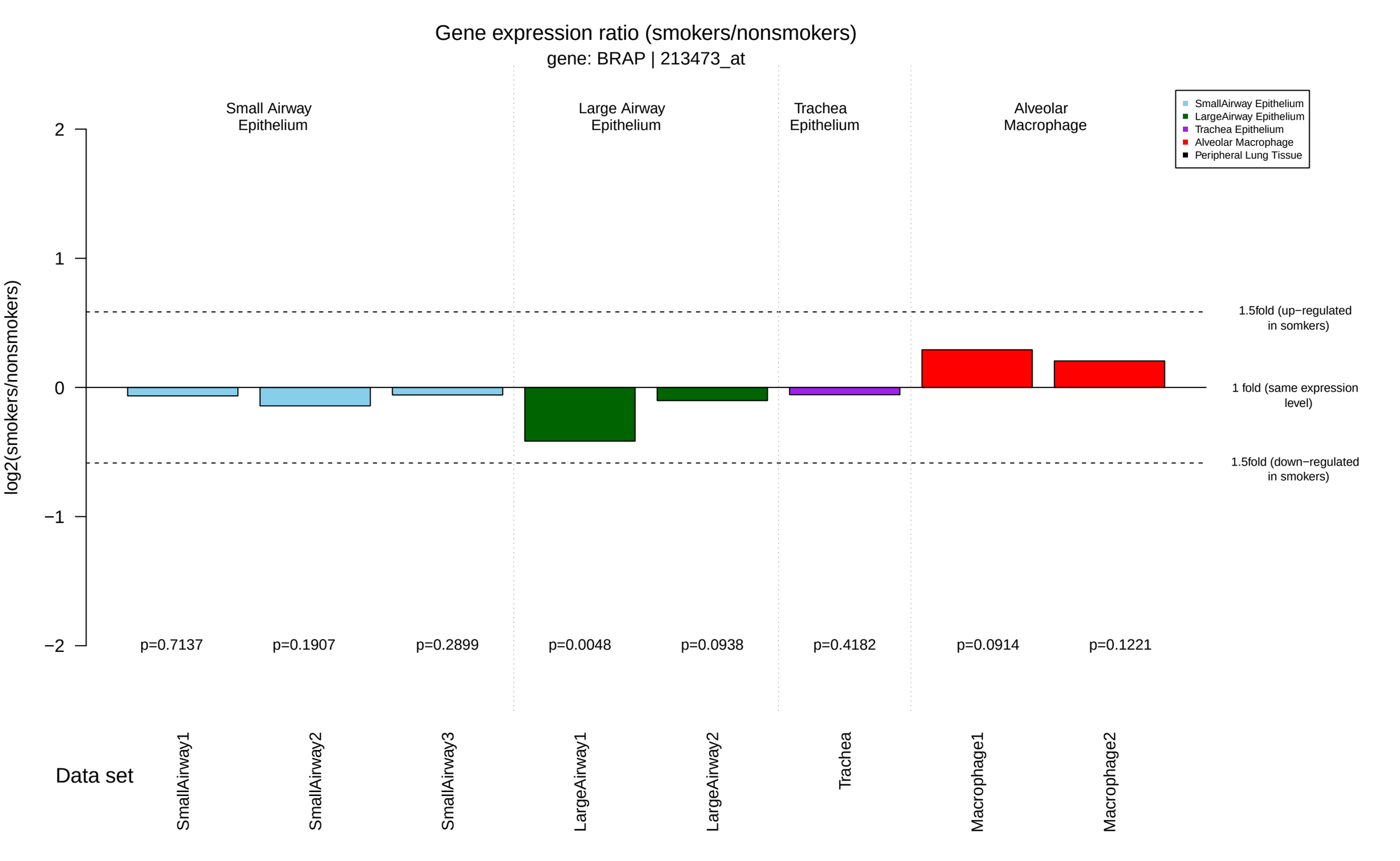
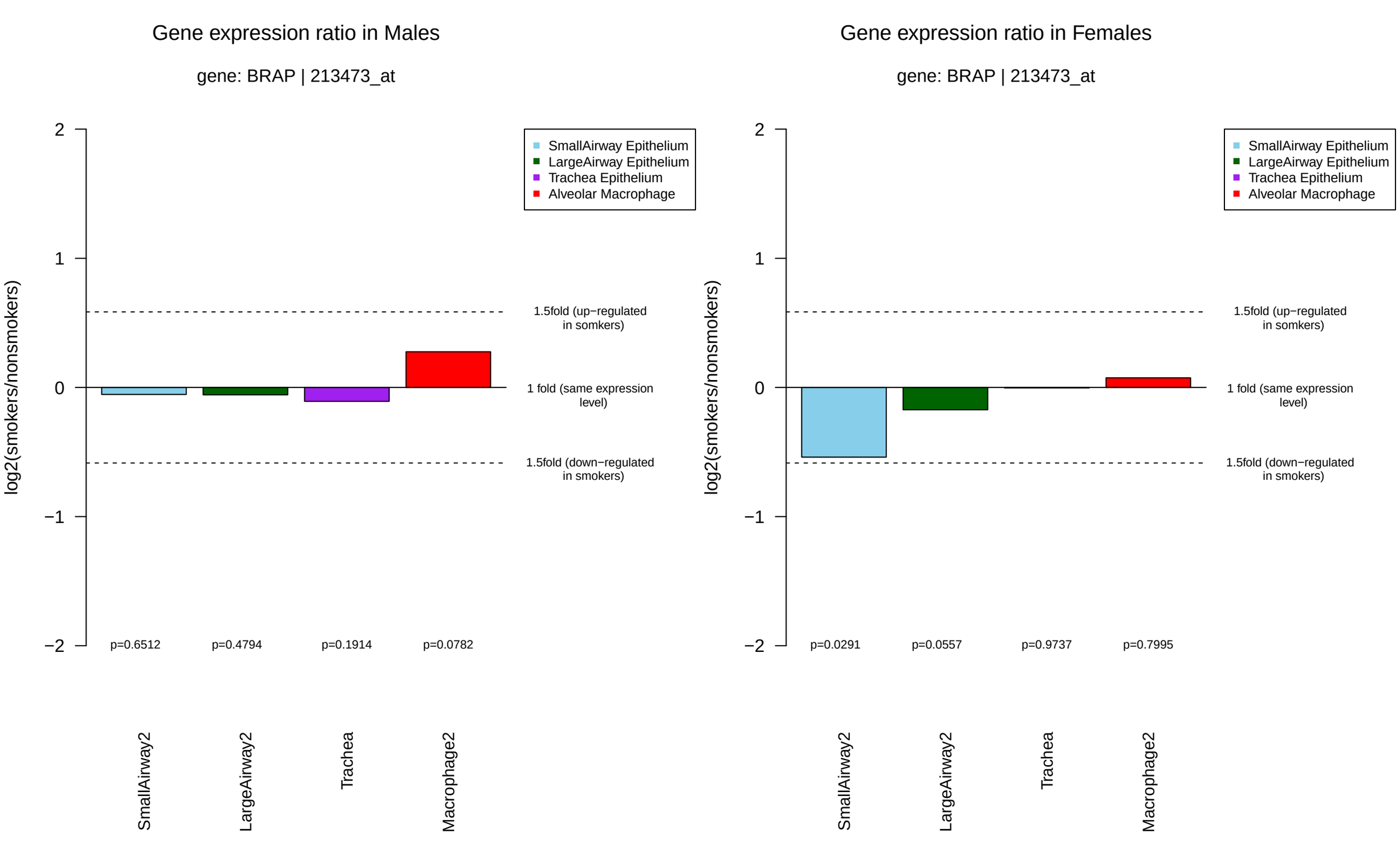
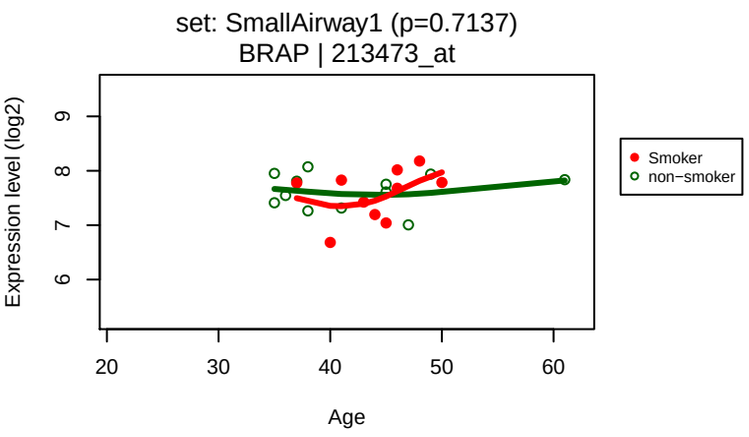
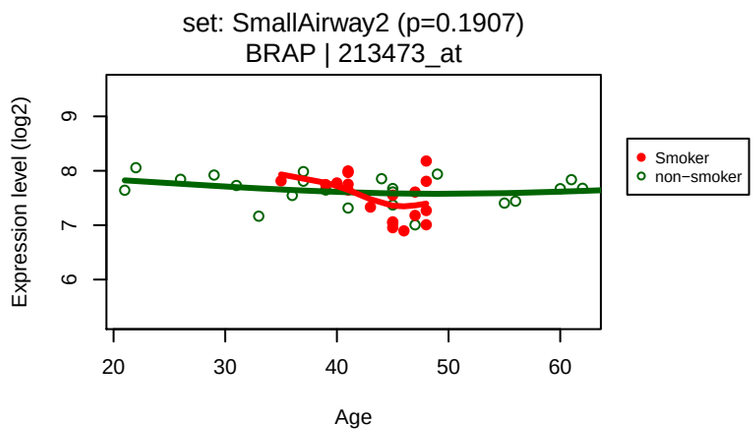
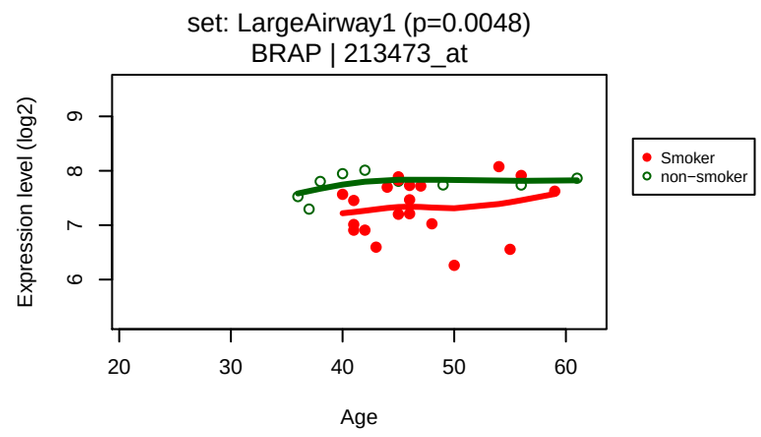

Gene expression of smokers and nonsmokers
gene: BRAP | 213473_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7137
p=0.1907
p=0.2899
p=0.0048
p=0.0938
p=0.4182
p=0.0914
p=0.1221
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

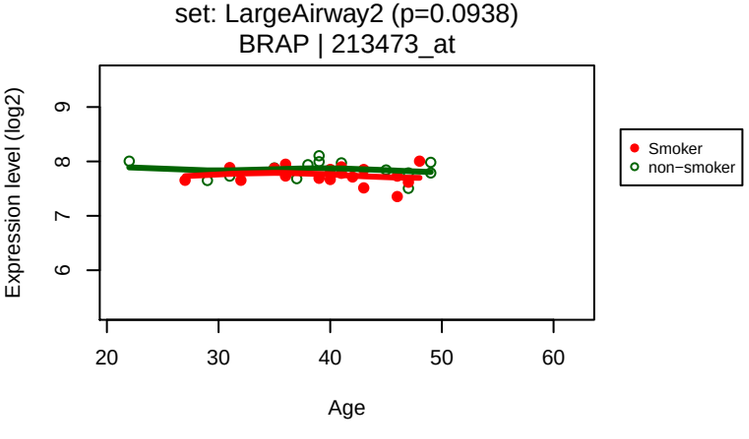

Gene expression ratio (smokers/nonsmokers)
gene: BRAP | 213473_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7137
p=0.1907
p=0.2899
p=0.0048
p=0.0938
p=0.4182
p=0.0914
p=0.1221
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

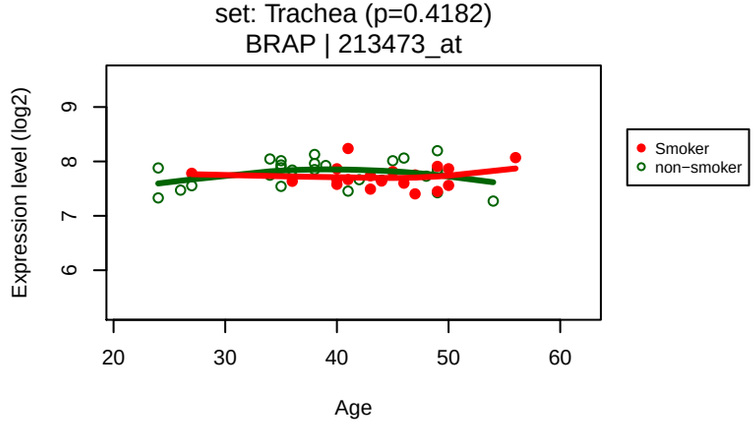

Gene expression ratio in Males
gene: BRAP | 213473_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6512
p=0.4794
p=0.1914
p=0.0782
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRAP | 213473_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0291
p=0.0557
p=0.9737
p=0.7995
SmallAirway2
LargeAirway2
Trachea
Macrophage2

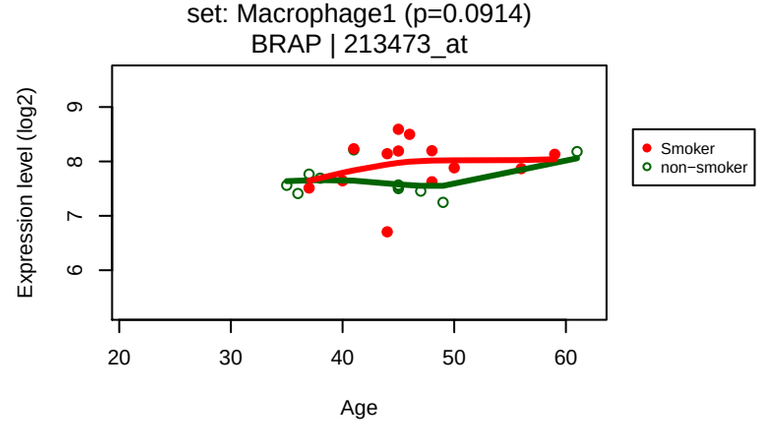

set: SmallAirway1 (p=0.7137)
BRAP | 213473_at
Expression level (log2)
Age
Smoker
non−smoker

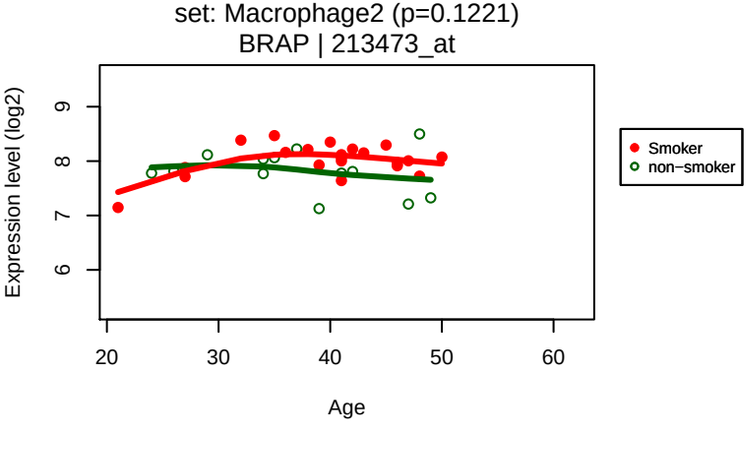

set: SmallAirway2 (p=0.1907)
BRAP | 213473_at
Expression level (log2)
Age
Smoker
non−smoker

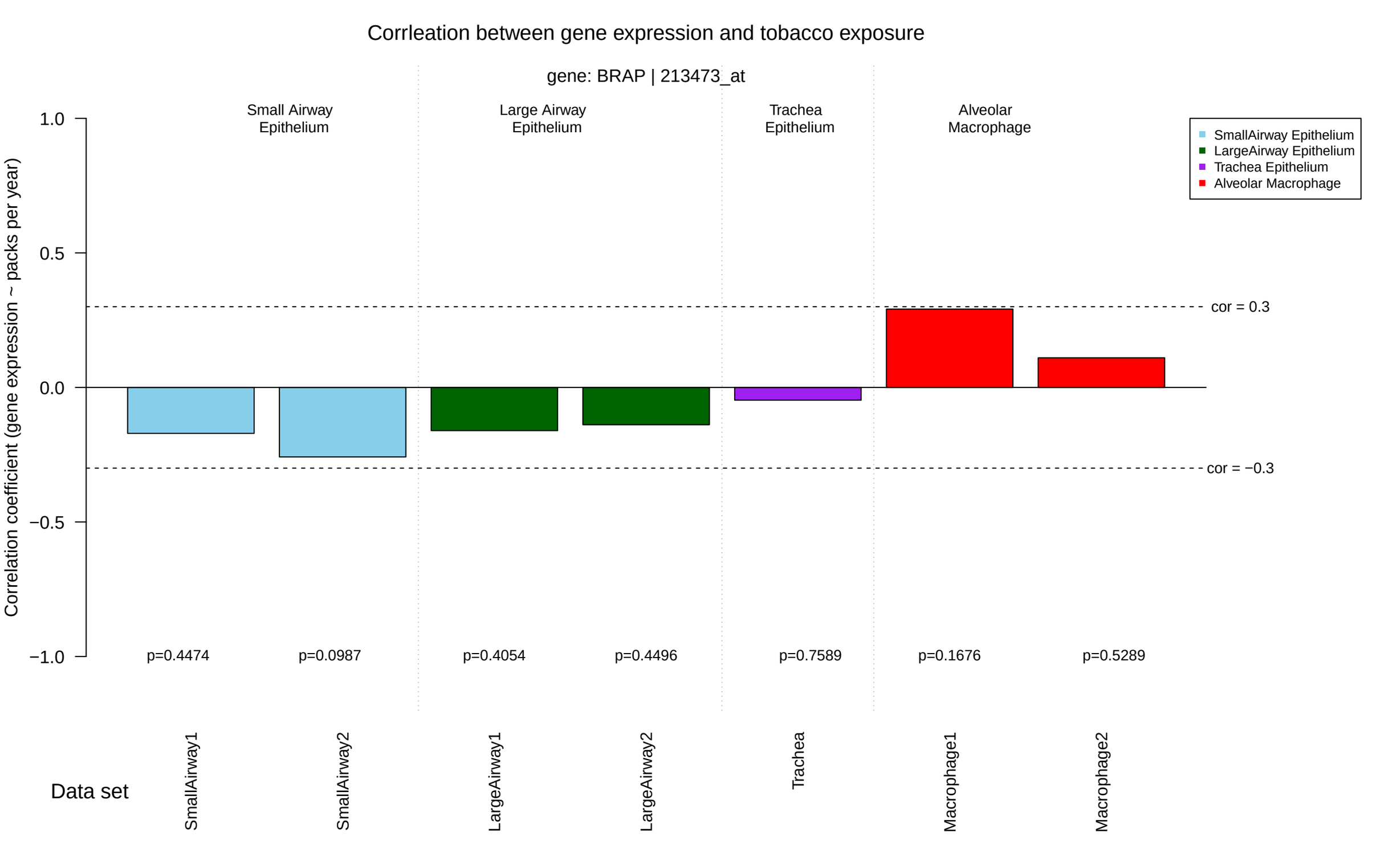

set: LargeAirway1 (p=0.0048)
BRAP | 213473_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0938)
BRAP | 213473_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4182)
BRAP | 213473_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0914)
BRAP | 213473_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1221)
BRAP | 213473_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRAP | 213473_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4474
p=0.0987
p=0.4054
p=0.4496
p=0.7589
p=0.1676
p=0.5289
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2